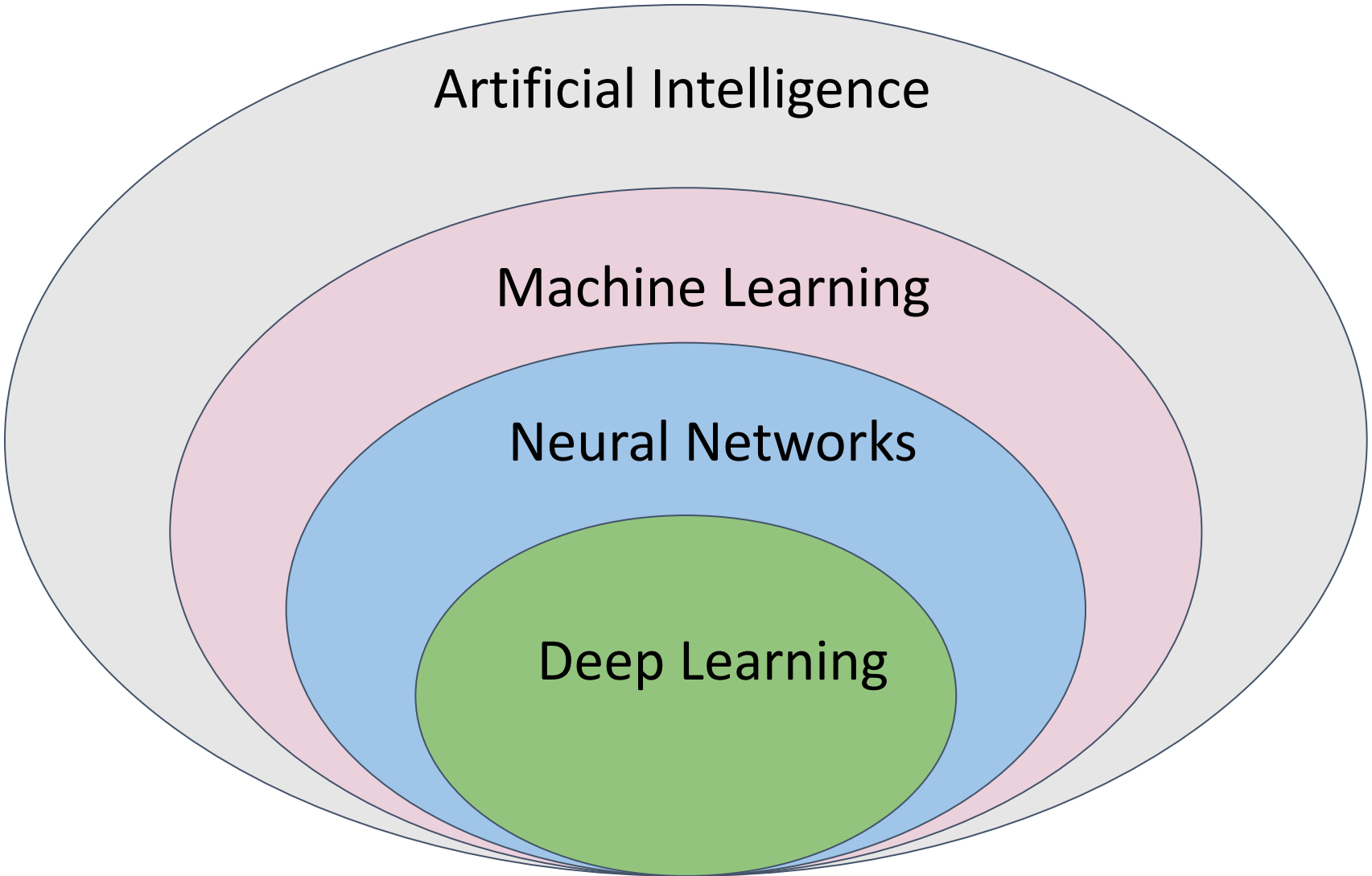
Introduction to Machine Learning

Dept. of CSE, University of Asia Pacific

Learning

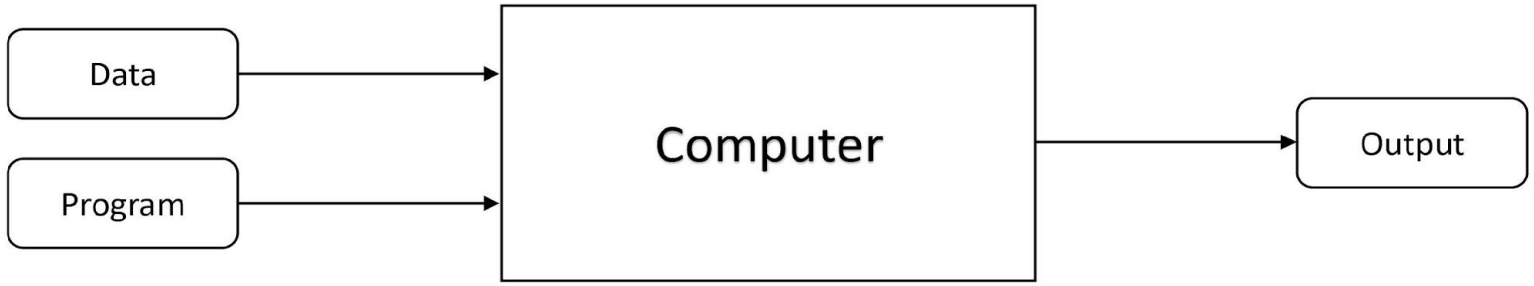
- Humans learn from past experiences
- Machines follow instructions
- What if humans can train machines to learn from past experience?
- Machines are like toddlers

Realm of Artificial Intelligence

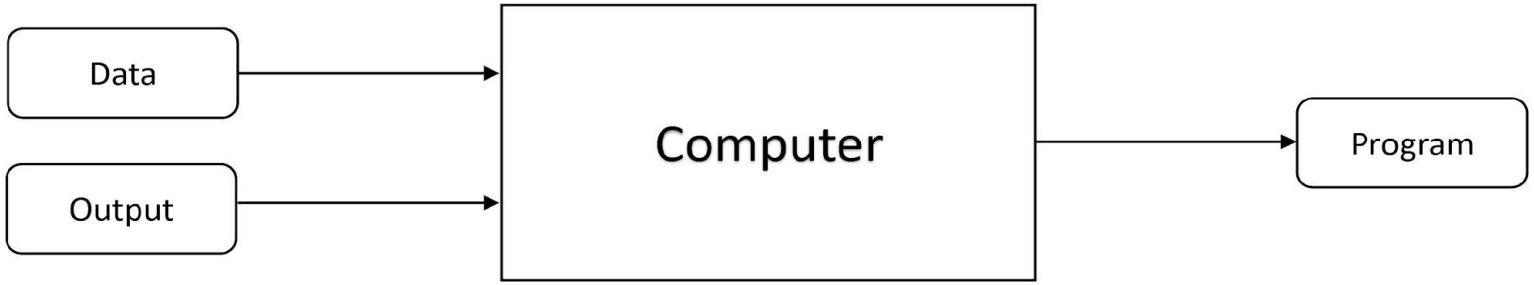


Programming vs Machine Learning

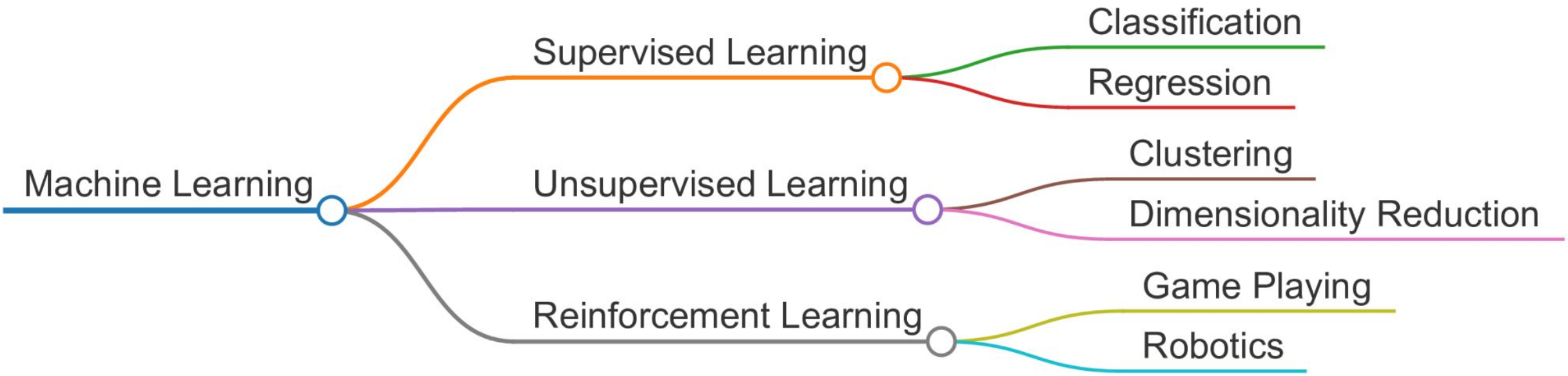
Traditional Programming



Machine Learning



Types of Machine Learning



Supervised Learning

A type of learning where the model is trained on labeled data.

Examples:

- Classification (e.g. Logistic Regression, Support Vector Machines)
- Regression (e.g. Linear Regression)

Unsupervised Learning

A type of learning where the model works with unlabeled data to discover patterns.

Examples:

- Clustering (e.g., K-Means)
- Dimensionality Reduction (e.g., PCA)

Reinforcement Learning

A type of learning where an agent learns to make decisions by interacting with an environment.

Examples:

- Autonomous Robot Navigation
- Self driving cars

General Steps in Machine Learning

1. Data Collection
2. Data Preprocessing
3. Model Development
4. Model Training
5. Model Evaluation
6. Deployment

Questions?